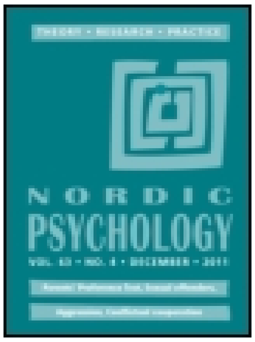

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RESEARCH ARTICLE

Blind spots in visible learning: A critique of John Hattie as an educational theorist

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Abstract

In recent years, John Hattie's book *Visible Learning* (2009) has greatly influenced educational practitioners and policymakers. The visible learning approach has been deemed "the Holy Grail of teaching" (Mansell, 2008), and Hattie has been called the "messiah" of educational research (Evans, 2012). In this article, we outline some of the significant methodological problems embedded in Hattie's work and relate them to his theoretical stance. We argue that his focus on single causal factors causes him to disregard important dimensions in educational practice. Furthermore, by analyzing parts of the primary research and the meta-analysis upon which Hattie grounds his conclusions, we find both serious methodological challenges and validity problems. We relate these problems to the technological rationality that informs Hattie's work and implicitly constitutes his theoretical approach. Finally, we outline, among other things, how questions of human agency and intentionality for attending school become marginalized as the broader consequences of using Hattie's approach to institutionally organize teaching processes. Another consequence of Hattie's work is that educational research begins with questions of methods rather than research into schools' everyday teaching practices.

Keywords: methodological problems, theory, educational status, Hattie

In writing his books on the subject of visible learning (e.g., Hattie, 2009, 2012; Hattie & Yates, 2014), John Hattie became a significant voice in the contemporary educational debate on teaching and learning. Mansell (2008) called Hattie's book *Visible Learning* (2009) "the Holy Grail of teaching." Hattie's apparently high-science approach, with structured quantitative data, thousands of research studies, effect sizes, and simple answers to complex problems were exactly what education policymakers and practitioners worldwide had sought (Zhao, 2018). Evans (2012) wrote, "He is not the messiah, but for many policy makers he comes close. John Hattie, possibly the world's most influential education academic, has the ear of governments everywhere." In the Danish context, educational researchers have called his works "the Hattie revolution" (Qvortrup, 2015), and others have presented Hattie as the researcher par excellence upon whose results the teachers in Danish primary schools can (and should) build their teaching. Skeptics are warned: "People who refuse to use Hattie's and others' results accept a substantial moral responsibility" (Hansen et al., 2015, p. 7).

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Hattie characterizes his work on "visible learning and teaching" as a synthesis. It is the interpretation of a very large amount of meta-analyses (at the time of writing, 1,200) based on and divided into a number of single effects (at the time of writing, 195).[1] In the original book *Visible Learning* (Hattie, 2009), he analyzed more than 800 meta-analysis building on 52,000 individual studies. In the chapter "Bringing It All Together," Hattie argued that the purpose of his project was not only to provide empirical data but also to construct a model for what characterizes good learning and teaching. Hattie (2009) wrote: "I aimed to build a model based on the theme of 'visible teaching, visible learning' that not only synthesized existing literature but also permitted a new perspective on that literature" (p. 237).

It is often emphasized that Hattie's conclusions are thoroughly grounded in unparalleled empirical evidence (Polanin et al., 2017), and a frequent general comment is that Hattie's work simply presents undeniable facts; this is also how Hattie understands himself and his project. In an interview, Hattie presented himself as "a measurement researcher; I am a statistician, I am not a theoretician"[2] (Knudsen, 2017, p. 259). Like his research, Hattie presents himself as a scientist who simply states the facts without the influence of theory.

The main points of this paper are that (a) Hattie's empirical work is far less solid than characterized, both by Hattie and those who wish to implement his visible learning platform in educational practice, and (b) that Hattie's research is both empirical and theoretical, and Hattie should thus also be understood as an educational theorist. In this article, we outline some of the significant methodological problems embedded in Hattie's work and relate them to his theoretical stance. We argue that his focus on single causal factors leads him to disregard important dimensions in educational practice. Furthermore, parts of the primary research and the meta-analysis upon which Hattie grounded his conclusions have serious methodological challenges and validity problems. We relate these problems to the technological rationality that informs Hattie's work and implicitly constitutes his theoretical approach. Finally, we outline some of the broader consequences of using Hattie's approach to institutionally organize teaching processes.

Hattie's findings are not the be-all and end-all of educational questions. When we claim that Hattie is a theorist, we argue that Hattie's theoretical approach has clear implications for his empirical research—for what he considers to be data, how he interprets data, and what he includes in and excludes from his empirical research. Furthermore, Hattie's theoretical stance creates a number of empirical problems within his work. That is, Hattie's reading of his empirical data may be better understood as an interpretation of educational practice through the perspective of his theoretical models rather than simply a matter of outlining and explaining the empirical data itself. This separates our contribution to the discussion of Hattie's work from those that developed a statistical critique of Hattie's work alone (e.g., Bergeron & Rivard, 2017; Simpson, 2017; Topphol, 2012) as well as those that discussed the theoretical implications in their own right. For example, Rømer (2019) claimed that Hattie's theory should be understood as a radical constructivist theory, following the footsteps of Glazerfeld, Biggs, and Bjørgen. We understand Hattie's theory differently and draw different conclusions from Hattie's implicit theoretical considerations than Rømer (2019) did. Hattie's major theoretical contribution lies in the technical worldview embedded in his empirical material and in the consequences this has for educational practice in abolishing notions of autonomy and meaning, as well as expanding educational fragmentation. Importantly, we are not discussing meta-analyses per se but the theoretical assumptions in Hattie's work, which used meta-analysis. It is important to underline that we are not critical of quantitative research within educational research in principle, either. We are critical of empirical research (both quantitative and qualitative) that lacks sufficient quality—that which does not make its theoretical assumptions transparent. Nor do we suggest that Hattie himself believes that he has the full account of what good teaching might be; indeed, he explicitly stated the opposite (Hattie, 2009). For instance, he readily admitted that his research is limited to achievement and that there may well be a discrepancy between the suggested interventions when they are part of a research project and when they are implemented by regular teachers (Hattie, 2009). More specifically, Hattie was also transparent about being interested in main effects and writes that "there are far fewer moderators than are commonly thought" (Hattie, 2009, p. 10). Hattie's primary interest in main effects is often contradicted when he interprets certain effects. Hattie himself mentioned a number of shortcomings of his own work, but he and those who find his research groundbreaking seem to forget these caveats when writing suggestions for educational practice.

Implicit theory

Theoretically, we claim that Hattie' work is strongly inspired by what Schön (1983, 1987) termed *technical rationality*, which is

> the heritage of Positivism, the powerful philosophical doctrine that grew up in the nineteenth century as an account of the rise of science and technology and as a social movement aimed at applying the achievements of science and technology to the well-being of mankind. (Schön, 1983, p. 31)

The engineer's design and analysis of materials and artifacts as well as the physician's diagnosis and treatment of disease have become the model of science-based technical practice in education. The growth of positivism within the social sciences has distorted the ideals of natural science upon which it was originally founded, and it has developed into an independent technical rationality far from how research within the natural sciences is conducted in practice (Kvale, 1973; MacIntyre, 1984). Technical rationality paves the way for means–end thinking in the social sciences, in which problems can be understood as entities in themselves beyond their context and can be solved by consulting and applying knowledge from basic science. Hattie's theoretical foundation in technical rationality is evident in how he is inspired by behaviorism. Hattie (1987) clarified that he drew on Glaser's theories in particular for his empirical work in education. In this context, Glaser's theory of learning is interesting. Hattie (1987) described Glaser's theory of learning as a "unique contribution" in connection with Glaser's model's focus on "the transformation process between the initial state and a state of competence" (Resnick, 1989, p. 191). Glaser's theoretical work was strongly inspired by B. F. Skinner's radical behaviorism. Glaser was a student of Skinner and is known for further developing Skinner's theories of programmed instruction and behavioral analyses into professional teaching designs (Resnick, 1989, xiv). Hattie's theoretical approach bears superficial similarities to Skinner's work in terms of its systematic attempts to identify conditions that reinforce the frequency of particular behaviors[3] as well as its reflection of Skinner's skepticism toward using theory when conducting scientific research.

MacIntyre (1984) made a similar point regarding the social sciences' ambitions of becoming objective sciences. In MacIntyre's analysis, the social sciences, in the pursuit of being truly scientific, developed the idea that empirical data lend themselves to direct interpretation without the need for a theoretical framework. Empirical investigations produce facts that allow for generalizations and ultimately use those generalizations as a basis for interventions. From this perspective, Hattie's somewhat ambivalent relation to theory is connected to the idea that although a "story" might be needed to communicate the findings, the numbers (i.e., generalizations of facts) are what count as the real science.

From our perspective, the issue is not that empirical research is inspired by theory. On the contrary, all research is theoretical in some respects when it comes to defining the problem, the questions asked, and the methods employed as well as in the interpretation of the results. Every theory involves a reduction of its subject matter. The problem, however, occurs when researchers do not explicate how their empirical work is related to and based on theoretical assumptions. Failing to elucidate one's theoretical assumptions creates the implicit impression that the results flow unmediated from the empirical data while appearing objective and atheoretical. One consequence of this is that others have propagated Hattie's findings as being "the only" answer to current educational woes, and without need for discussion. This trend is highly problematic. In this paper, we outline how Hattie's work is embedded in a particular theoretical perspective and how this fact makes a difference in interpreting Hattie's empirical work.

Empirical problems

By focusing on how single inputs (i.e., distinct instructional strategies) are related to student achievement when measured quantitatively, Hattie's work followed a central premise in technological rationalism. Manipulating instructional strategies as external variables makes it possible to identify the functional responses of manipulations (i.e., effects/students' learning). By using visual barometers in relation to every theme, Hattie imparted the impression of clear effect relations and presented clear-cut and identifiable causes that necessarily lead to particular effects.[4] The recipients of Hattie's texts certainly read his barometers/effect sizes as causal effects. In the Danish context, Hattie's outlines of single causes—such as class sizes and the impacts of homework—and their effects have had direct implications for educational practice. The results have been used to legitimize increasing class sizes in the Danish educational system, and some high schools have abandoned homework (Carboni, 2013). In effect, Hattie's work made fundamental statements about the educational situation: First, it is possible to identify key parameters and directly determine how they are functionally related to changes in the rate of responses within this framework; second, it is possible and meaningful to isolate extraneous variables (O'Donohue & Kitchener, 1998). The theoretical assumptions embedded in Hattie's empirical approach tend to atomize educational practice, legitimizing addressing each intervention separately rather than comprehending educational practice as interrelated, with different interventions dynamically influencing each other in various ways.

In pursuit of single causal factors in a complex world

Hattie pursued the effects of discrete causal factors (e.g., homework, class size, or feedback). We aim to discuss how (or whether) it is meaningful to delimit single causal factors within a complex setting (e.g., an educational environment). Furthermore, Hattie himself did not discuss how he identified the causal factors when measuring given effects. This strategy became problematic for Hattie with respect to class size.

According to Hattie (2009), the effects of teaching children in small classes relative to large classes are limited (p. 86), effect of class size $d = 0.21$. However, upon taking a closer look, this conclusion has several problems. Berger (1982) emphasized that using class size as an analytical unit in itself makes no sense when assessing its effects on student performance.[5] Based on analysis of the existing literature, Berger (1982) opined that class size could not be understood as a meaningful analytical unit in itself. Instead, class size should be understood in relation to students' abilities and social background, heterogeneity in the class, the teacher's ability to teach, and the teacher's resources, among other things. Another problem arises when scrutinizing the school's educational reality: how should class size be operationalized? In many schools, the students are in different classes or work in groups with varying numbers of participants throughout the school day (Berger, 1982). Therefore, it can be difficult to determine when a student is part of a small class or a larger class. Furthermore, in educational practice, schools use class size strategically, such as by making class sizes smaller to help challenge children academically or giving teachers more time to work personally with the youngest children.

Despite Hattie's finding that class size is actually a meaningful analytical causal factor with which to work, he considered other explanations as to why class size does not seem important in relation to pupil performance, including whether the circumstances of the teaching situation are different in large versus small classes. Hattie (2009) himself pointed out that teaching in larger classes (30–80 students) probably requires a skilled teacher and that teaching in such a context is highly structured, with a focus on overlearning of basic skills. Thus, the teaching performed in different studies is actually different and therefore very difficult to compare. The point here is not to discuss whether Hattie's assumptions are solid, but to point out that he himself described the difficulty in deciding what drives the findings. The crucial variables in relation to the effect being measured are the teaching in classes with few pupils, the allocation of skilled teachers to larger classes, and whether there are few or many pupils in the different classes. As Hattie asserted that important differences may exist between teaching in differently sized classes and because these considerations are not incorporated into his analysis, his indication of a single effect size for class size, as such, must be considered problematic. In some cases, it can be legitimate to work with single-factor designs; however, the problem is that this is the only strategy Hattie applies in his approach to educational research.

The elimination of moderation—social economic status as an example

The internal logic of a given effect size is matched by another omission that Hattie made by ignoring the complexity of everyday life and a number of other variables that may affect the outcomes of his calculations. Danish statistician Allerup (2015) argued that examining individual variables in isolation within an educational context leads to a problematic reduction in complexity. Instead, according to Allerup (2015), it is specifically important to work with multiple variables simultaneously (multivariable analyses) to uncover these complexities. When attempting to use statistical analyses of individual elements, special care must be taken to demonstrate, rather than posit, that single elements have individual effects.

In Hattie's pursuit of an evidence-oriented perspective to education, he selected single causal relations in isolation as his point of departure. This principle has important implications. One is that Hattie deliberately eliminated factors that others researching the same general phenomena included in their analyses. A good example of this is Hattie's treatment of pupils' socioeconomic backgrounds, which he chose to omit as a moderator of results in other areas. It could be argued that by eliminating socioeconomic status from his analysis, Hattie omitted social class issues from his analysis. This separation of the pupils' social and economic background factors from ideas of teaching seems a political act in itself.[6] According to Hattie, teachers cannot counter students' socioeconomic background factors in school. Instead of including socioeconomic factors in analyses of other educational phenomena—for instance, feedback—Hattie (2009) calculated a separate effect size for socioeconomic background, $d = 0.57$. Hattie (2009) legitimized this choice by arguing that he would only deal with educational matters that could be influenced through teaching (p. x–xi). However, it is generally acknowledged in educational research that pupils' socioeconomic background is a central key to understanding teaching's impact in schools today (Snook et al., 2009). In principle, it is difficult to say what percentage of pupils' learning is affected by their socioeconomic background or in what ways learning is affected by it. However, it doubtlessly plays an important role. The most solid finding in an Organisation for Economic Co-operation and Development (2005) report was that the most important source of variation in the students' learning, in relation to both their attitudes and skills, was their social and family backgrounds (Blichfeldt, 2011; Snook et al., 2009).

Hattie's choice to exclude socioeconomic status had implications for what is happening in schools—pupils have socioeconomic backgrounds that influence their performance. From the schools' perspective, Hattie was essentially excluding social background and social class from discussions regarding teaching practice when taking this tabula rasa perspective of pupils. Hattie might have been right when claiming that teachers cannot change pupils' socioeconomic backgrounds; however, pupils' socioeconomic backgrounds undoubtedly have a massive impact on everyday teaching practice, and it is this influence that Hattie took out of the equation. By neglecting the importance of pupils' socioeconomic background, Hattie's work, at worst, developed class-blind pedagogical thinking.

The production of a significant effect

In his analysis, Hattie focused on effect sizes—specifically those that were pragmatically significant and made a marked difference in everyday student learning. Educators should scrutinize these kinds of effect sizes, as Hattie used some of these significant effect sizes in his synthesis and in his model for good teaching. The choices that Hattie made regarding whether effect sizes were significant—to some degree based on subjective choices that Hattie himself made—resulted in important consequences.

For each of the original 138 factors that Hattie (2009) examined, he calculated an effect size as described above. Almost all interventions in the study (95% according to Hattie, 2009, p. 15) had positive effects—almost everything worked. This premise gave Hattie the opportunity to introduce a cutoff score for when an effect size is great enough for its implementation to be relevant. Hattie decided that the cutoff would be $d = 0.40$ based on pragmatic logic:

> This cut was the average effect from 800-plus meta-analyses and the book fully elaborates on the origin of this claim. *It is a threshold to create a story* (about what is common, above and below this effect) but it is not presented as a fixed cut-score below which one should ignore and above which one should cherish—many parts of the book go to some length to show the importance of not making such conclusions. (Hattie, 2010, p. 89; emphasis added)

Whether Hattie himself thought that the above statement made a difference, he stated that the 0.4 threshold "provides a 'standard' from which to judge the effects" (Hattie, 2009, p. 17) and, further, that it "is the point on the continuum that provides the hinge or fulcrum around which all other effects are interpreted" (p. 16).

By setting the cutoff score at 0.4, Hattie turned his data into value statements. Although effect sizes themselves are already judgments about which strategies are more valuable and effective, a cutoff score is first and foremost a supplementary rhetorical or narrative tool that allows for the quick separation of the single factors upon which one should focus amid others. Application of a cutoff score turns data into a binary list of what can and cannot be considered educationally valuable. However, this mechanical outlining of what is significant and what is not educationally desirable is problematic. To decide what is valuable, one must focus on context and content. Mechanically deciding a universally fixed standard has important consequences. For example, the effect of giving people with heart problems low doses of aspirin to prevent heart problems is $d = 0.07$, which means that such a treatment saves 34 out of 1,000 people's lives (Snook et al., 2009). Based on Hattie's cutoff score, this is a low effect, but in terms of saving 34 people's lives, it may be a worthwhile action, considering that the aspirin treatment is low cost, has few side effects, and is easy to adopt. Hattie's cutoff score of 0.4 is not a scientifically "true" or objective categorization of when an intervention should be considered significant. This is a highly normative question.

In many respects, Hattie's account of the many distinct factors developed what Wrigley (2018) termed a "closed system," in which each factor is separated from one another and from the environment in general. According to Wrigley (2018), the educational practice of everyday life is characterized as a result of dynamic interactions among both external and internal school factors. As argued by Wrigley (2018), the closed systems depicted in Hattie's empirical work are characterized by ideas of causality, eliminating ideas of human agency and intentionality.

Hattie's effects hide important dimensions

Once a total effect size has been calculated, it can very well hide considerable differences. For example, the effect of homework shows how Hattie essentially hid important dimensions of what he wanted to clarify by using a cutoff score of $d = 0.4$. Hattie found that the effect of giving the pupils homework was $d = 0.29$, which fell below his chosen cutoff score. This effect size has been publicized as if homework in no way influences pupils' learning (Snook et al., 2009, p. 94), as Hattie himself asserted in an interview: "'Traditional homework has a vanishingly small effect, and it's getting worse the younger the students are', explains the international celebrity researcher John Hattie."

A number of educational institutions in Denmark have followed Hattie's results and have given up on homework (Carboni, 2013). However, this conclusion is problematic. Despite what the above quote suggests, the evidence does not show that homework is ineffectual. On the contrary, the effect of homework in elementary/primary school is 0.15 (clearly below the cutoff score of 0.4), while its effect in high school is 0.64 (clearly over the cutoff score; Snook et al., 2009). The results thus appear to be that homework actually has a significant impact on high school and upper-secondary school learning, but this insight disappears when only a weighted general average of homework's importance is outlined as 0.29 (Hattie, 2009, p. 234). Hattie (2009) acknowledged this variability: "There may be many moderators. For example, the typical effect size of homework is $d = 0.29$, but the effects are greater for high school students and closer to zero for elementary school students" (p. 17). However, the cutoff score is not just higher than 0.4 for students at the high school level, but considerably higher. Additionally, it matters whether the students are academically strong or weak. The importance of homework in different subject matter (e.g., mathematics in relation to subjects such as Danish and social studies) also disappears in Hattie's generalization (see Hattie, 2009, pp. 234–236). The point is that, in some cases, Hattie's reduction of voluminous empirical material to discrete effect sizes obscures important differences rather than making them more visible.[7]

The above reveals an important feature of how Hattie treats the content of many important meta-analyses: namely, as generalizations based on a very general definition of a phenomenon (homework is homework, regardless of the pupils' age, the subjects in which it is given, etc.) and apparently with little concern about the loss of information that results from such generalization.

Feedback as an example of educational effects

Hattie claimed that the high number of studies involved in substantiating his claim make the effect sizes outlined in his meta-analysis reliable. In many respects, this essential claim makes Hattie's work so convincing: The many studies supporting the significant effect sizes make the results more reliable. However, this claim's problems include presupposing that Hattie's meta-analyses follow the same rigid methodological procedures and essentially all research the same phenomenon; otherwise, it makes no sense to compare and add them together.

In the following, we use feedback as an example to argue that Hattie's meta-analysis was based on very different studies (for an elaborate analysis of the feedback concept, see Ekecrantz, 2015). Therefore, as outlined below, we question whether Hattie's effect size for feedback came from studies that scrutinized the same phenomena. Notably, a crucial part of Hattie's synthesis model of teaching relies on the importance of feedback, again showing his close affiliation with behaviorism, which sees reinforcement's equivalency to feedback as a significant concept. One could argue that Hattie's understanding of feedback becomes educationally impoverished because of his strong affiliation with behaviorism (for a more comprehensive account of feedback, see Shute, 2008). Feedback belongs at the top of Hattie's ranking of single factors (Number 10, with $d = 0.73$) and is central to Hattie's argumentation for a theory of visible learning and teaching. Before publishing *Visible Learning*, Hattie published "The Power of Feedback" in 2007 with a colleague to indicate the importance Hattie places on feedback (Hattie & Timperley, 2007). This applies to the feedback pupils receive on their work and the feedback teachers receive from pupils about their perspectives and challenges.

In *Visible Learning*, Hattie (2009) defined feedback as "information provided by an agent (e.g., teacher, peer, book, parent, or one's own experience) about aspects of one's performance or understanding" (p. 174). He asserted that "feedback is a 'consequence' of performance" (Hattie, 2009, p. 174). In an educational context, feedback is the key to learning and can be understood in this context:

> In summary, feedback is what happens second, is one of the most powerful influences on learning, occurs too rarely, and needs to be more fully researched by qualitatively and quantitatively investigating how feedback works in the classroom and learning process. (Hattie, 2009, p. 178)

Importantly, even though Hattie professed great faith in feedback's educational power—especially by focusing on information—researchers who studied feedback remain skeptical. Through other meta-analyses and reviews of feedback, these researchers have a more ambiguous assessment of feedback's impact. In a review of formative feedback—and in agreement with continuing work by Kluger and DeNisi (1996)Shute (2008) concluded the following: "Within this large body of feedback research, there are many conflicting findings and no consistent pattern of results" (p. 153). Thus, in contrast to Hattie's assumptions about homogeneous research on feedback, Shute concluded that research in feedback is far from unambiguous.

We will explore the disagreement regarding feedback's effects by studying five meta-analyses upon which Hattie based his conclusions. In Hattie's original book on visible learning (2009), his synthesis about feedback comprised 23 meta-analyses with 67,931 people. The effect of using feedback strategies was $d = 0.73$—a very high effect. Among the 23 meta-analyses Hattie referenced in his synthesis of feedback research, five particularly heavy analyses included 62,761 people, equivalent to 92% of the total sample. In the following, we examine the quality of these five meta-analyses that Hattie used to understand feedback: Kluger and DeNisi (1996); Lysakowski and Walberg (1980); Lysakowski and Walberg (1982); Swanson and Lussier (2001); and Witt et al. (2006). In the following, we briefly discuss these five meta-analyses in greater detail.

Lysakowski and Walberg (1980)

Lysakowski and Walberg (1980) meta-analysis included 4,842 participants and a mean effect of 1.17 (Hattie, 2009, Appendix A). The meta-analysis did not consider feedback the way Hattie (2009) defined it. Lysakowski and Walberg (1980) did not mention feedback; instead, they focused on reinforcement techniques: "In the studies, 39 distinct reinforcement techniques were identified that were used either to initiate or to improve educational behaviour, and generally to enhance educational outcomes of students in learning settings" (Lysakowski & Walberg, 1980, p. 115). Although Lysakowski and Walberg (1980) did not initially give a clear definition of reinforcement, they did in their 1982 paper. There, reinforcement included "the extent to which the student is rewarded for his learning" (Lysakowski &

Walberg, 1982, pp. 560–561). In Hattie's definition, he explicitly critiqued reinforcement by arguing that "it is doubtful whether rewards should be thought of as feedback at all" (Hattie, 2009, p. 174). Even though Hattie dismissed a central dimension of reinforcement, he included a meta-analysis that focused on reinforcement techniques. Furthermore, information played a central role in Hattie's definition of feedback. This differs from Lysakowski and Walberg (1982), who analyzed classical reinforcement strategies' effects in the classroom.

Lysakowski and Walberg (1982)

Lysakowski and Walberg (1982) meta-analysis included 15,689 participants and a mean effect of 0.97 (Hattie, 2009, Appendix A). The meta-analysis focused on three dimensions: cues, participation, and corrective reinforcement. One central problem of the article is its focus on instructional strategies rather than on feedback, namely cues and participation (Lysakowski & Walberg, 1982, pp. 560–561).

The problems increase after evaluating the research studies upon which Lysakowski and Walberg based their own meta-analysis. Looking at two studies (from a list of 54) referenced by Lysakowski and Walberg (1982), Wentling (1973), and Travers et al. (1964) clarifies the various conceptions of feedback and under which conditions it is measured.

Wentling's (1973) and Travers et al. (1964) studies are nearly entirely different. Wentling (1973) evaluated how feedback was specifically applied during a 5-week general automotive mechanics course in a high school. Travers et al. (1964) simulated a school situation involving four feedback conditions: fourth-, fifth-, and sixth-grade students were instructed to learn 60 German words over 3 consecutive days. Wentling (1973) and Travers et al. (1964) researched feedback in different contexts (high school vs. an experiment) with different participants (male high school students vs. fourth to sixth graders) and while working with different subjects (learning to repair motors vs. learning a language). However, the largest problem lies with the experiments' focus on different aspects of feedback: Wentling (1973) focused on missing, partial, or full responses on a test, but Travers et al. (1964) focused on interactions related to feedback on a test. Whether either study supports Hattie's definition of feedback, which focuses on feedback's information, remains questionable. Wentling (1973) found that receiving partial feedback was effective for the students, but Travers et al. (1964) found that verbal interaction provided effective feedback. In both cases, one can argue that the context in which the feedback is delivered partially determines whether it is effective.

Kluger and DeNisi (1996)

Kluger and DeNisi (1996) meta-analysis included 12,652 participants and a mean effect of 0.38 (Hattie, 2009, Appendix A). Only in Kluger and DeNisi (1996) meta-analysis do we find a feedback definition similar to Hattie's: "This article is about FIs [feedback interventions] defined as actions taken by (an) external agent(s) to provide information regarding some aspect(s) of one's task performance" (p. 255). Interestingly, Kluger and DeNisi (1996), in a study which Hattie (2009) described as "the most systematic study addressing the effects of various types of feedback" (p. 175), concluded that feedback interventions have an effect size of $d = 0.38$—much lower than the 0.73 that Hattie reached. In addition, 38% of the studies included in

Kluger and DeNisi (1996) work had a negative effect on the learning process, which contradicts Hattie's assumption that "almost everything works". Contrary to Hattie's largely positive approach to the "power of feedback," Kluger and DeNisi (1996) described feedback as something that effects performances significantly more or significantly less.

Witt et al. (2004)[8]

Witt et al. (2004) meta-analysis included 24,474 participants and a mean effect of 1.15 (Hattie, 2009, Appendix A). According to Hattie's synthesis (2009), this study included the most participants and showed the second-greatest positive effect for feedback issues. In other words, Hattie claimed this study substantially helps to emphasize his conclusions about feedback. Besides the validity problem (addressed below), the study contained additional problems. Many research studies supporting the meta-analysis are inaccessible: 11 are unpublished doctoral dissertations, five are unpublished master's theses, and six are paper presentations from various conferences. These 22 studies include over 7,539 participants.

Regarding validity (i.e., whether the study measures what it aims to measure), the article did not address feedback issues directly; rather, it focused on the importance of teachers' *immediacy*. The text does not mention feedback. In other words, the notion of immediacy as feedback's proximity in time is central to Witt et al. (2004) meta-analysis; however, this aspect is not central to Hattie's definition of feedback, which focused on information rather than immediacy. How immediacy relates to Hattie's understanding of feedback remains unclear: "This meta-analysis provides a systematic quantitative summarization of the studies of verbal and nonverbal immediacy of teachers in relation to students' affective, cognitive, and perceived learning outcome" (Witt et al., 2004, p. 150). Hattie's definition of feedback sees immediacy as, at most, a part of the feedback process and information as the essential part of feedback.

More problems arise upon examining three studies (Chesebro & McCroskey, 2001; Frymier & Houser, 2000; Gorham & Christophel, 1990) from the work of Witt et al. (2006). None of these three studies explicitly concern feedback, and they all focused on immediacy. Teachers' immediacy constitutes only one part of these three studies. They also focus on receiver apprehension (Chesebro & McCroskey, 2001), communication skills (Frymier & Houser, 2000), and humor (Gorham & Christophel, 1990). None of the three use particular strategies for selecting participants or conduct interventions, and they only measure immediacy via self-reported questionnaires. The poor fit of these studies raises the issue for meta-analyses coined by Eyshenk as "garbage in, garbage out," in which fundamental methodological problems in the primary studies carry over to further meta-analyses, making problems difficult to identify (Borenstein et al., 2009, p. 380). The quality of meta-analyses depends on the quality of the included studies, so a meta-analysis that includes studies with a low degree of rigidity will undermine the whole study.

Swanson and Lussier (2001)

Swanson and Lussier (2001) meta-analysis included 5,104 participants and a mean effect of 1.12. The meta-analysis focused on the role of examiner assistance but not on feedback as defined by Hattie:

> One possible alternative or supplement to traditional assessment is to measure a child's performance when given examiner assistance. A procedure that attempts to modify performance, through examiner assistance, in an effort to understand learning potential, is called dynamic assessment (DA) … . When a student is having difficulty, the examiner attempts to move the student from failure to success by modifying the format, providing more trials, providing information on successful strategies, or offering increasingly more direct cues, hints, or prompts. The intensity of the intervention ranges from several sessions to brief intensive prompts in one session. Thus, potential for learning new information (or accessing previously presented information) is measured in terms of the distance, difference between, or change from unassisted performance to a performance level with assistance. (Swanson & Lussier, 2001, pp. 321–322)

In this excerpt, Swanson and Lussier (2001) revealed their focus on examiner assistance. However, they did not address information from a feedback context because the feedback allowed in an examination situation only includes "cues, hints or prompts" (Swanson & Lussier, 2001, pp. 321–322).

Summary

In four of the five heaviest meta-analyses connected with Hattie's coverage of feedback, it is unclear whether they used a feedback term identical to Hattie's (see also Blichfeldt, 2011). Two of the five meta-analyses did not even mention feedback (Lysakowski & Walberg, 1980; Witt et al., 2004). This realization raises a question: Did the large number of meta-analyses used by Hattie research the same phenomenon (feedback), or did he subsume different phenomena under a single header? Assuming the latter, we must consider Hattie's conclusions as invalid. Similarly, our analysis indicates low transparency in two of the five studies. In addition, only one study consistently used a control-group design. In general, Hattie (2009) does not give the reader any account, criteria, or justifications for how he uses meta-analyses. As indicated above, Hattie could just as well have used other meta-analyses in the synthesis when it comes to feedback.

Discussion

This article initially asserts that Hattie built his analysis from a philosophy of technological rationality. Above, we found problems in Hattie's apparently factual and statistical stance on educational research as outlined in his famous work *Visible Learning* (2009). In searching for single causal factors, he reduced educational complexity (e.g., class size), eliminated important moderators (e.g., social economic status), constructed a threshold for what counts as educationally desirable out of his concern to tell a good story (e.g., a threshold of 0.40), divided the material so that important dimensions and nuances disappear (e.g., homework), and piled various studies measuring various phenomena together under the same heading for the sake of persuasion. We argue that the connection Hattie makes to a theory of technical rationality has the consequences that a dialogue about education, teaching, and learning becomes too narrow. As a consequence, his ideas of visible learning become narrow as well, risking that all dialogues about education, teaching, and learning become obsolete.

Further, the lack of dialogue results from the lack of conceptualization of human agency and intentionality. Hattie (2009) gave no concept of teachers' or students' interests, no concept of intentionality, and no understanding of the school or the role that school plays in students' lives. His work does not portray students or teachers as seeking or interpreting meaning. Hattie only discussed students' experiences in relation to how students may be susceptible to the teacher's influence. Although students must be active, they are not initially understood as acting intentionally. Students only act mechanically as a consequence of feedback. The students' intentionality and uniqueness are replaced by an understanding of their effectiveness in solving problems. Instead of addressing students as intentional agents, Hattie's work discusses, for example, self-efficacy, self-handicapping, self-motivation, self-goals, and self-dependence (Hattie, 2012; Hattie & Yates, 2014). These concepts did not come from Hattie's own empirical work but from various theories, primarily from constructivism and cognitive psychology. Hattie used all of these concepts as replacements for students' interpretations of education's relevance and importance, and these concepts serve Hattie's ambition of helping students become their own teachers. The understanding starts not at understanding the teaching situation or the school from the students' perspective but in finding aspects of student activities that could be corrected. Thus, students maintain their image as problem solvers.

Hattie addressed educational issues from the school's and the teacher's points of view (and from a particular school's and teacher's understanding), with students as objects under the school's or teacher's influence, rather than addressing school issues by understanding the school's role in students' lives. Furthermore, Hattie's approach to education seems to replace human dialogue with a strong focus on mutual feedback processes between the teacher and student, and he provided little explanation of why students should learn certain things.

Another argument claims that Hattie's overall aim included real effects on how teachers (and students) participate. He apparently hoped that his list of effect sizes (and barometers), along with the visible learning theory's guidelines, would sufficiently secure educational progress. Even with some problems in Hattie's inclusive stance on researching student achievement, a new set of problems appeared when Hattie insisted, contrary to the original fields (e.g., regarding feedback), on reducing any concept like this to single-effect sizes.

Paradoxically, Hattie attempted to synthesize an educational practice that he has already broken down (into single-cause effects) without concern for due diligence in calculating whether an assumption holds for the individual effects he calculated. Due to Hattie's theoretical assumptions, his research developed what Wrigley (2018) termed a closed system, with little reference to everyday educational practice. The educational practices facing pupils, teachers, parents, and educational planners every morning constitute an open system in which each educational intervention relates to other interventions and in which what happens in school fits into what happens outside the classroom. In many respects, Hattie's theoretical assumptions construct educational practice as a separate universe, with few connections to schools' everyday practice. In our view, studying how to improve teaching should begin by researching schools' everyday teaching practices as well as researching the meaning that pupils, teachers, and parents attach to participating in education. Regarding Hattie's work, research should not begin by considering what methods to use to improve modern schooling. Playing on the visible learning metaphor, we can ask if Hattie actually produced a fata morgana with his research. The central question seems to concern for whom he produced the fata morgana. Eacott (2017) provided an interesting answer, viewing Hattie's work as a version of neo-Taylorism that provides school administrators and school leaders with a scientific language for legitimizing changes within educational school systems worldwide. However, we believe the theoretical blind spots in Hattie's work make modern educational systems' paradoxes even greater, not smaller, and that we must address the theoretical assumptions embedded in educational research.

Excluding human agency and intentionality is arguably problematic when addressing human learning and development. German philosopher Honneth (1996) stressed the importance of recognizing persons' agency and intentionality as crucial for their development as autonomous participants (Nielsen, 2016). Conversely, Honneth (2008) claimed that failing to acknowledge or recognize autonomy and intentionality could cause social alienation. In Hattie's pursuit of theoretical, empirical, and practical certainty, he confounded errors and weaknesses, thus damaging his intention to create a theory supported by the widest base of empirical evidence supporting teachers' daily practices.

Notes

1. See www.visiblelearningmetax.com for an update.
2. Bergeron (2017), in a paper dedicated to critiquing Hattie's use of statistics, concluded, "In summary, it is clear that John Hattie and his team have neither the knowledge nor the competencies required to conduct valid statistical analyses" (p. 245).
3. Hattie did not directly address how his work was influenced by behaviorism; however, within certain themes such as feedback, he discussed how the concept of feedback is contrary to Skinner's positive reinforcement principle. He wrote that his understanding of feedback and information "is not the same as a behavioral input–output model" (Hattie & Timperley, 2007, p. 82).
4. By applying a meta-analytical approach, Hattie created a methodological backdoor for himself concerning the inference of causality from his analysis:

   Often I may have slipped and made or inferred causality—and in some cases reasonably so. Certainly, the fundamental word in meta-analysis, effect size, implies causation (What is the effect of a on b?) and this claim is often not defensible. The claims in this book are more oriented to developing an explanation—a plausible set of claims based on evidence. It is more an abductive than an inductive or deductive exercise (Haig, 2005)—the explanation or story offers a plausible theory, a set of inferences to the best explanation in light of my experience of reviewing and interpreting the many studies, and it is hoped the story is bold enough to be potentially disprovable. My task is to present a series of claims that have high explanatory value, with many (refutable) conjectures. (Hattie, 2009, p. 237)

   Contrast this with Simpson's (2017, 2018) refutation of the assumption that effect sizes can be used as reliable indicators of effects.
5. For more recent discussions of class size, see Blatchford and Russell (2019) and Mathis (2017).
6. Bergeron (2017) made a succinct point: Hattie tends to calculate effects so that they yield positive numbers. Regarding socioeconomic status, the value of 0.59 was exactly that: positive. However, it may as well have been calculated as a negative effect (affecting the least privileged students) of −0.59. Bergeron (2017) clarified, "negative impact is considered [by Hattie] to be harmful."
7. Gene Glass, from whom the idea of meta-analysis originated, warned sharply against using meta-analysis to cloud the variation and heterogeneity of scientific data (Wrigley, 2018, p. 367).
8. Hattie's book *Visible Learning* references Witt et al.'s (2004, 2006) meta-analysis. The number of participants, studies, and effect sizes that Hattie referenced in his Appendix A fit the version published in *Communication Monograph* (Witt et al., 2004). However, Hattie consistently referred to the version published in a monograph (Witt et al., 2006). We see this as a formal mistake and thus referred to the 2004 version in this article.

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